

JONATHAN SANSO

Full-Stack / AI Engineer | LangChain/LangGraph, On-prem LLM & RAG, Face Recognition, OCR/NLP/STT | Multi-tenant SaaS (Supabase RLS)

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SUMMARY

Full-Stack Developer with **4 years** of experience building web products end-to-end (multi-tenant SaaS, internal platforms, AI services). Core expertise in **tenant isolation** (Supabase RLS/Policies), **POS/orders lifecycle**, **RBAC**, **AI engineering** (LLM orchestration with **LangChain/LangGraph**, offline **RAG**, face recognition embeddings, OCR/NLP/STT), **legacy DB migrations**, and **security-first architecture** (MFA, CSRF, HMAC/JWT). Daily stack: **Next.js**, **React**, **TypeScript**, Tailwind, **Supabase/PostgreSQL**, **NestJS**, Prisma, **Python** (FastAPI, Pandas, **InsightFace**). Experienced with **Docker**, **NGINX**, **AWS**, Terraform, **Redis/BullMQ**, **SendGrid**, Mercado Pago API.

Core Competencies: Full-Stack Development | Software Engineering | AI Engineering | LangChain | LangGraph | LLM Orchestration | Offline RAG | LLM Evaluation | On-premise / Air-gapped AI | Vector Search | Face Recognition | OCR | Speech-to-Text | Multi-tenant SaaS | RBAC | Row-Level Security | REST APIs | Microservices | CI/CD | Legacy DB Migration | ETL | Agile / Scrum | Code Review

TECHNICAL SKILLS

Frontend: Next.js, React, TypeScript, JavaScript (ES6+), Tailwind CSS, HTML5, CSS3, Zustand, TanStack Query, Zod, react-hook-form, @dnd-kit, ShadCN UI

Backend: NestJS, Node.js, Prisma, FastAPI (Python), Express, GraphQL, REST APIs / RESTful, OpenAPI / Swagger, Microservices

Databases: PostgreSQL (Supabase RLS, Triggers, RPC, Indexes), MySQL, MariaDB, MongoDB, SQL Server, iBase (legacy migration)

DevOps & Infra: Docker, NGINX, Debian/Linux, AWS (EC2, RDS, S3, CloudFront), Terraform, CI/CD, GitHub Actions, Vercel, Git/GitHub, PM2

AI & Data: LangChain (LCEL), LangGraph, Ollama / Qwen2.5, local embeddings, vector store on MariaDB, structured output (JSON Schema), Pydantic guardrails, golden-set evals, Gemini API, InsightFace (512-d face embeddings), Tesseract OCR, faster-whisper STT, NLP, Pandas, NumPy, jsPDF, ExcelJS

Integrations: SendGrid (transactional email), Mercado Pago, Redis/BullMQ, Webhooks (HMAC signed)

Testing & Methods: Unit Testing, Jest, integration tests, Agile / Scrum, Kanban, Code Review, Git Flow

Security: Auth0, MFA/2FA, CSRF, Helmet/CORS, HMAC-SHA256, JWT, Rate Limiting, Input Sanitization

PROFESSIONAL EXPERIENCE

Ocean Stack -- Full-Stack Developer / Software Engineer (Multi-tenant SaaS) - Dec 2025 - Jun 2026

Products: Niappa POS | Oceans HR (ATS) [oceansstack.com](#)

- Shipped **weekly product increments** in a structured delivery loop: **Figma** as source of truth, daily alignment with **CTO + PM**, iterative releases, and **UX/UI polishing** (Next.js, TypeScript, Tailwind).
- Implemented **tenant-safe isolation** and **role-based permissions** across modules so each business only sees and mutates its own data (**Supabase Postgres**, **RLS/Policies**).
- Delivered **operational back-office flows** (products, ingredients, stock, counts, suppliers) with **strong data consistency** guarantees via DB automation (**Postgres triggers**, indexes, **SQL/RPC**).
- Implemented **orders at scale: table sessions + split checks**, resilient **order lifecycle**, and safe **edit/void/restore** flows keeping stock accurate end-to-end (**RPC**, **transactional integrity**, **idempotent stock adjustments**).
- Built **reporting exports** for operations: **PDF/Excel** (sales + cash/audit), plus productivity-focused UI improvements (**i18n ES/EN**, search, sorting, confirmations).
- Built **Oceans HR ATS module** serving **hundreds of CVs/month per tenant** across multiple active clients: **Kanban pipeline with drag-and-drop**, vacancy management, candidate profiles with notes/comments, **recruitment funnel reports**, **RBAC with company-level roles**, full **ES/EN i18n** (Next.js 16, React 19, TanStack Query v5, Zod v4, @dnd-kit, Supabase RLS).
- Engineered **AI-Powered CV Matching Engine**: **LLM integration** (Gemini API with **3-model cascade fallback** for HA), **PDF text extraction**, **weighted scoring across 8 criteria** (skills, seniority, role, education, location, salary, language, industry), **structured JSON output**, plus full frontend UX with drag-and-drop upload, color-coded score cards, per-criteria progress bars, and **automated candidate creation pipeline** (Next.js App Router, Supabase RLS).
- Implemented **SendGrid transactional email pipeline** triggered by **candidate pipeline phase changes** (**automated** and **semi-automated flows** with recruiter approval step): **templated emails per stage**, **Supabase triggers + queued dispatch**, **idempotency keys**, and **delivery audit log per tenant**.

Argentine Federal Penitentiary Service -- Full-Stack Developer / AI Engineer / Software Engineer - Jan 2024 - Present

Private repos - Sige2i internal platform (NestJS + Next.js + Python/FastAPI)

- Shipped an **on-prem NL intelligence chatbot** over a **90+ table legacy database: SQL-RAG** (deterministic parameterized retrieval, **zero hallucinated facts**, zero data egress) + **grounded 100% local LLM** (Ollama/Qwen2.5, CPU-only, air-gapped), orchestrated with **LangChain LCEL** - **structured output** via server-side JSON Schema, **token streaming**, custom observability callbacks, **byte-for-byte verified parity**, feature-flagged rollout with **zero-downtime rollback**; inline **[n] citations** to source records; cut a >45s legacy self-join to **sub-second** via indexed single-entity queries + app-side intersection (Next.js, NestJS, Python/FastAPI, Prisma, MariaDB).
- Built a **LangGraph** orchestration layer with **semantic validation** and **feedback-driven retries** (**best-attempt strategy** - a retry never regresses below the prior result) plus a **map-reduce pipeline with parallel fan-out** per entity to beat small local models' context-window limits; backed by a **labeled golden set** and **A/B/C evaluation** across LLM backends, choosing the production default **from measured data**.
- Led delivery of a **two-platform ecosystem** (internal ops + public verification portal) with **secure data flows** and controlled access for multiple stakeholders (React, Next.js, TypeScript, Tailwind, NestJS, Prisma).
- Built **Python Face Matcher service** in production: **InsightFace buffalo_I** (ONNX CPU) **512-d L2-normalized embeddings**, in-memory (N, 512) float32 index with **NumPy cosine brute-force + argpartition top-K**, **per-UID centroid** (mean + renormalize) for multi-photo enrollment, persisted as **BLOBs in MariaDB**. Endpoints **/face/embed**, **/face/search**, **/face/search-multi** (**FastAPI**).
- Built an **AI-assisted document intelligence pipeline**: **OCR** (Tesseract LSTM spa+eng with grayscale + binarization threshold 140), **hybrid PDF extraction** (PyPDF2 for native text, pdf2image + Poppler @ 300 DPI fallback with per-page OCR, NDJSON streaming progress), and **ASR** (**faster-whisper small INT8 CPU** normalizing WhatsApp .opus via ffmpeg), all with **human-in-the-loop review**.
- Migrated **~110 GB of legacy iBase8 data** to **MySQL/MariaDB** across **80+ modules**: relational tables plus heterogeneous binary content (**PDFs**, **scanned images**, **ZIP archives**, **Word documents**). Designed **normalized relational schema**, built **ETL pipelines** with chunked streaming and resumable runs, **classified and stored binary assets** with checksum integrity, validated data integrity end-to-end, and integrated migrated entities into the new **NestJS + Next.js** stack.
- Automated **operational reporting** (**PDF/Excel/Word**) and engineered **secure document verification** (**time-limited access**, **cryptographic validation**, **HMAC-SHA256**, **JWT**, watermarking) replacing manual workflows (TypeScript, Pandas).
- Implemented **end-to-end security** (MFA, CSRF, rate limiting, **Helmet/CORS**) and **Redis/BullMQ** async job queues with in-memory fallback caching, on **containerized infrastructure** for **170 concurrent users** (**Docker**, **NGINX**, **PM2**, **Debian**) with automated backups.

VirtuaState -- Frontend Developer - May 2022 - Dec 2023

- Built a responsive marketing/portfolio site for a VR/AR studio with **SEO** and performance optimizations (HTML5, CSS3, JavaScript, mobile-first).
Live: <https://www.virtuastate.net/>

EDUCATION

Higher Degree in Programming - Teclab - Aug 2022 - Aug 2024

- Software fundamentals and delivery practices: databases, web development, UX, cloud basics, project management.

Full-Stack Web Development - CoderHouse - Jan 2022 - Mar 2023

- Delivered multiple end-to-end projects: frontend and backend foundations, web app architecture (HTML5, EJS, CSS/SASS, React, Node.js, MongoDB, Express, GraphQL).
Live: <https://react-e-commerce-j-sanso.vercel.app/>